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Introduction to Machine Learning
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"Automated Paper Reviewing with a Compact LLM: LoRA Fine-Tuning"

Abstract

This paper assesses whether a compact ~3 B-parameter language model, adapted with QLoRA, can support peer review by generating structured feedback and an explicit Accept/Reject decision. Training uses OpenReview ICLR 2017–2019 data, with editorial decisions mapped to a binary label and a token-level cross-entropy optimised over review text plus decision. Evaluation is strictly out of distribution on ICLR 2020. A two-prompt zero-shot baseline (Llama-3.2-3B-Instruct) is fluent but severely biased, accepting 2 199 of 2 203 submissions and achieving only 30.9 % accuracy. Fine-tuning with QLoRA updates about 1.33 % of the parameters and shifts the decision distribution to ≈ 1 650 Accept / 546 Reject (N ≈ 2 196), raising accuracy to 41.7 %. Beyond accuracy, the tuned model improves lexical overlap with human reviews (e.g. token-Jaccard mean 0.130→0.146, Weaknesses ROUGE-1 0.149→0.169, Strengths chrF 0.205→0.217) while reducing validation loss and perplexity (≈ 2.27 and ≈ 9.7 vs. ≈ 3.0 and 20 for the backbone).
The zero-shot baseline retains a small advantage on semantic similarity metrics (Sentence-BERT precision/recall/F1) and on Strengths-ROUGE-L.
A KL-regularisation experiment, aimed at inheriting the baseline's fluency and semantic alignment, over-constrained adaptation and degraded loss and perplexity without improving accuracy. Overall, lightweight QLoRA adaptation moves a compact model from "fluency first" toward "fluency with selectivity," offering a practical, reproducible improvement for early-stage peer-review support.

Introduction

Peer review is the primary mechanism for quality control in scholarly publishing, yet it is expensive, slow and often inconsistent. Instruction-tuned language models can generate fluent prose on demand; however, fluency alone is insufficient for this task—an automated reviewer must also render a binary verdict. In this study I explore whether a compact ~3 B-parameter LLM, adapted with LoRA/QLoRA, can deliver a disciplined first pass on submissions by producing a concise, structured critique and an explicit Accept/Reject decision. Prior work has explored automated peer review from multiple angles. Yuan et al. (2022) asked whether NLP models could generate first-pass reviews for scientific papers, finding that generated texts were less constructive and less factual than human reviews, and concluding that the technology was not yet ready for high-stakes settings. More recently, Zhou et al. (2024) evaluated GPT-3.5 and GPT-4 on score prediction and review generation, finding that while LLMs can give passable decisions, they remain weak on zero-shot scoring and critical feedback—a limitation consistent with the extreme acceptance bias I observe in my zero-shot baseline. Idahl and Ahmadi (2024) introduced OpenReviewer, a system fine-tuned specifically on peer review data that achieves superior accuracy compared to general-purpose LLMs, demonstrating the value of domain adaptation. This work extends that direction by asking whether a similar benefit can be obtained with a compact 3B-parameter model via QLoRA, making the approach accessible without large-scale compute. To test genuine generalisation, I train only on OpenReview dumps from ICLR 2017–2019 and reserve ICLR 2020 as a strictly out-of-distribution test. Each training record (title, abstract, review, decision) is normalised and cleaned; I construct prompt–completion pairs where the prompt concatenates the paper's title and abstract, and the completion concatenates the review text and a final decision line (Decision: Accept/Reject). Decisions are binarised using a simple heuristic ("accept" → Accept, else Reject); the decision label is never included in the prompt to prevent leakage. The remainder of the paper describes the preprocessing pipeline, the fine-tuning setup, and a comprehensive evaluation showing that this lightweight adaptation shifts the model from near-universal acceptance toward a more balanced and lexically grounded decision-making behaviour.
The complete project material, including the source code, is available at the following link:
https://drive.google.com/drive/u/1/folders/1XQKbqUhGCq4b5yfLhBnxhAM3o07K3kSm

Data and Preprocessing

We build our dataset from OpenReview's "absolute_data" dumps. Conferences from 2017–2019 are used for training/validation, while ICLR 2020 is held out for out-of-distribution testing. Each raw record includes a paper title, abstract, one or more review texts and an editorial decision. We first normalise column names and remove rows with missing title, abstract, review or decision. For the training split, we then construct prompt–completion pairs suitable for language-model fine-tuning:
– The prompt is formed by concatenating the paper's title and its cleaned abstract, separated by two newlines (`title + "\n\n" + abstract`). Prefixes such as "Abstract:###" are stripped during cleaning.
– The completion consists of the full review text followed by a binary decision line, e.g. `Decision: Accept` or `Decision: Reject`. We binarise the original editorial decision using a simple heuristic: any string containing "accept" (case-insensitive) maps to Accept, and all others map to Reject. This step also strips prefixes like "Recommendation:###" or "Decision:###". After assembling these pairs, we remove exact duplicates, assign a `year` field, and save the resulting dataframe as a Parquet file (`train_17_18_19.parquet`). For the 2020 test set, we perform only the cleaning steps (normalise column names, strip whitespace, drop missing values

and duplicates) and save the cleaned table to `test_20.parquet`; ; we do not append the decision to the review, because the true label is used solely for evaluation and is never included in the model's prompt to avoid leakage. This pipeline ensures that the model trains on structured prompts closely matched to the format it will generate at inference time while keeping the out-of-distribution test data strictly separate.

Metrics

The task is a binary classification of the editorial decision (Accept vs. Reject) coupled with a generative review. The primary metrics are macro F1 and per-class precision/recall/F1, which provide a balanced view of model behaviour on the class-imbalanced test set; overall accuracy is reported alongside them but interpreted with caution, since a naive always-Reject classifier would achieve 69.1% accuracy by exploiting the class skew. We also inspect the predicted class distribution to detect degenerate behaviours (e.g., "always accept"). To evaluate review quality, I used lexical overlap metrics—Jaccard, ROUGE-1/2/L, and chrF—that quantify token/character n-gram overlap with human reviews, checking adherence to the requested structure and coverage of strengths/weaknesses. Because surface overlap can miss paraphrases, I reported semantic similarity—SBERT (cosine; precision/recall/F1) and BERTScore (F1)—to assess meaning-level alignment. Finally, I tracked cross-entropy and perplexity as language-model (regression-style) diagnostics of predictive uncertainty and fluency; they are not target metrics but help interpret the trade-off between fluency and the more balanced selectivity achieved by fine-tuning.

Baseline: Zero-Shot with Structured Prompting

As a reference system, I use Llama-3.2-3B-Instruct without any fine-tuning. Inference follows a two-prompt strategy. The first prompt asks the model to act as an ICLR peer reviewer and return a structured critique in a fixed format (Decision, Comment, Strengths, Weaknesses); the second prompt forces a single-token verdict (Accept or Reject) to guarantee a parseable binary label. Review generation uses light sampling (temperature 0.7, top-p 0.9, repetition penalty 1.15, no-repeat n-gram size 6, max 220 new tokens) to reduce repetition while maintaining fluency; the decision prompt uses greedy decoding for reproducibility. Despite always returning a decision, the baseline exhibits extreme acceptance bias: on the ICLR 2020 test set it predicts Accept for 2 199 of 2 203 papers, achieving only 30.9% accuracy. This behaviour is consistent with the instruction-tuned backbone defaulting to the most constructive, least confrontational response—a prior that fine-tuning must overcome.

QLoRA Fine-Tuning

The same Llama-3.2-3B-Instruct backbone is adapted using QLoRA, which combines 4-bit NF4 quantisation (double quantisation enabled, bfloat16 compute dtype) with lightweight low-rank adapters injected into all seven projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. The LoRA rank is set to $r = 16$ with scaling factor $\alpha = 32$ and dropout 0.05. This updates 24 313 856 parameters out of 1 827 777 536 total — approximately 1.33% of the model — while keeping the remaining weights frozen in 4-bit quantised form, making the approach practical on a single commodity GPU.

Training uses a token-level cross-entropy loss computed only over the completion tokens (the structured review plus the final `Decision:` line), with prompt tokens masked out via label = −100. This forces the model to learn to generate a review and commit to a decision rather than simply echoing the input. The optimiser is AdamW with learning rate 1×10^{-4}, weight decay 0.1, warmup ratio 0.03, batch size 4 per device, and gradient accumulation over 8 steps (effective batch size 32). The context window is capped at 1 408 tokens. Training runs for up to 10 epochs with early stopping (patience 3, evaluated every 200 steps); the best checkpoint is selected by validation loss on an 8% holdout of the 2017–2019 data. The optimal checkpoint appears around step 400, at which point validation loss reaches ≈ 2.27 (perplexity ≈ 9.7), compared with ≈ 3.0 and ≈ 20 for the untuned backbone.

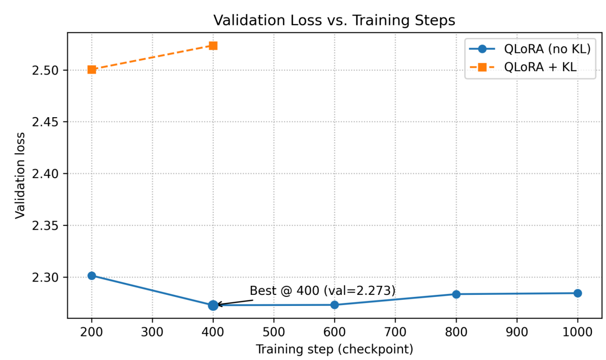


Figure 1. Validation loss vs. training steps. QLoRA peaks at step 400; KL underperforms.

Evaluation

All evaluation is performed on the held-out ICLR 2020 set, which is strictly out of distribution relative to the training corpus (ICLR 2017–2019). To avoid leakage, I reserve 8 % of the 2017–2019 data as a validation split and deduplicate items by *(title, abstract)*; during evaluation I match items using the same unique identifier. Editorial outcomes are mapped directly to a binary Accept/Reject label, and the model never sees the true decision during generation. The primary metrics are macro F1 and per-class precision/recall/F1 on the 2020 set; overall accuracy is reported alongside them but interpreted with caution given the class imbalance. As a sanity check against degenerate behaviour (e.g., "always accept"), I also monitor the distribution of predicted decisions (percentage of Accept vs. Reject). To assess the quality of the generated reviews beyond the final verdict, I compute a suite of text-similarity diagnostics between model outputs and human reviews. These include token-level Jaccard overlap, ROUGE-1/2/L and chrF scores on the strengths and weaknesses sections, as well as SBERT precision/recall/F1 and BERTScore (F1) to capture semantic alignment. These metrics provide complementary lexical and semantic perspectives on how closely the model's reviews mirror human feedback. Finally, I monitor validation loss and perplexity as regression-style signals of language modelling quality; although they are not used for selection, they help interpret the trade-off between fluency and selectivity observed in the results.

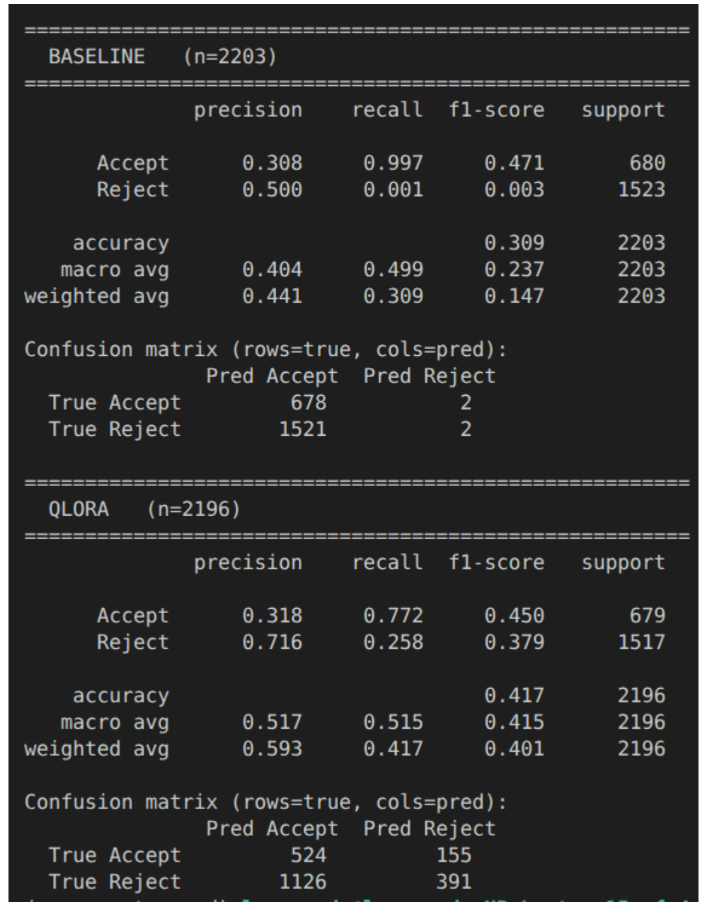
Results

On the ICLR 2020 test set (N ≈ 2 203), the zero-shot baseline predicts Accept for 2 199 papers and Reject for just 4, yielding an overall binary accuracy of 0.309. The QLoRA-tuned model outputs decisions for 2 196 papers with a much more balanced distribution — around 1 650 Accept and 546 Reject — raising accuracy to 0.417 (see Fig. 3–4). Table 1 summarises per-class precision, recall and F1 for both models (Fig. 2).

Before interpreting these numbers, a critical reference point is needed: a naive always-Reject classifier would achieve 69.1% accuracy on this test set, since 69% of submissions were rejected. Both models fall well below this ceiling, which confirms that overall accuracy alone is a misleading metric on class-imbalanced data. The macro F1 tells a more informative story: it rises from 0.237 (baseline) to 0.415 (QLoRA), reflecting genuine improvement across both classes rather than a shift in bias. The key contribution of fine-tuning is therefore not the absolute accuracy gain but the emergence of a functional Reject signal: Reject precision rises to 0.716, meaning that a paper flagged as Reject has roughly a 70% chance of matching the human decision — a credible first-pass filter where none existed before. The baseline fails on Reject because the instruction-tuned backbone defaults to the most constructive, least confrontational response; fine-tuning on ICLR data breaks this prior by exposing the model to the lexical and structural patterns that distinguish rejected submissions.

On review quality, QLoRA shows modest but consistent lexical improvements: mean token-Jaccard rises from 0.130 to 0.146, Weaknesses ROUGE-1 from 0.149 to 0.169, and Strengths chrF from 0.205 to 0.217. The baseline retains an advantage on semantic similarity (SBERT mean cosine 0.633 vs. 0.585) and on a few lexical scores such as Strengths ROUGE-L (0.124 → 0.110). This trade-off is consistent with the model shifting from fluent, abstract prose toward more domain-specific vocabulary: the gain in lexical specificity is useful for author feedback, while the loss in semantic closeness is a side effect of domain adaptation rather than a fundamental failure. On language-modelling signals, the fine-tuned model clearly dominates: validation loss drops from ≈ 3.0 to ≈ 2.27 and perplexity from ≈ 20 to ≈ 9.7; the baseline's edge lies solely in smoother fluency and higher semantic similarity.

This residual semantic gap motivated a KL-regularisation experiment aimed at inheriting the backbone's fluency while retaining QLoRA's selectivity. In practice the strategy backfired: the teacher distribution was itself strongly skewed toward Accept, so the KL penalty pulled the adapters back toward zero-shot behaviour, yielding higher validation loss (≈ 2.50 vs. 2.27) and worse perplexity (≈ 10.1 vs. 9.7). Decision accuracy was not evaluated for the KL-anchored model on the full ICLR 2020 test set due to computational constraints; the comparison is therefore limited to validation loss and perplexity, which already show clear degradation relative to vanilla QLoRA. Given that the primary goal of KL regularisation was to improve semantic similarity without sacrificing selectivity, and that neither objective was achieved, this negative result is sufficient to rule out this direction without a full inference run. A more balanced teacher distribution or an annealed KL schedule might fare better, but in the present setting the penalty hindered progress precisely when the model needed freedom to diverge from the backbone's semantics.



BASELINE (n=2203)

	precision	recall	f1-score	support
Accept	0.308	0.997	0.471	680
Reject	0.500	0.001	0.003	1523
accuracy			0.309	2203
macro avg	0.404	0.499	0.237	2203
weighted avg	0.441	0.309	0.147	2203

Confusion matrix (rows=true, cols=pred):

	Pred Accept	Pred Reject
True Accept	678	2
True Reject	1521	2

QLORA (n=2196)

	precision	recall	f1-score	support
Accept	0.318	0.772	0.450	679
Reject	0.716	0.258	0.379	1517
accuracy			0.417	2196
macro avg	0.517	0.515	0.415	2196
weighted avg	0.593	0.417	0.401	2196

Confusion matrix (rows=true, cols=pred):

	Pred Accept	Pred Reject
True Accept	524	155
True Reject	1126	391

Figure 2. confusion matrix + metrics

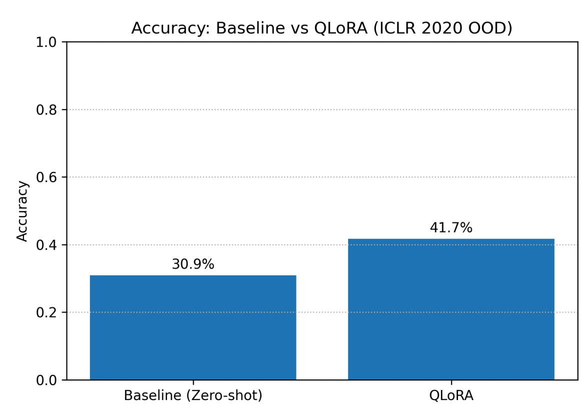


Figure 3. Accuracy on ICLR 2020 (out-of-distribution): zero-shot baseline vs. QLoRA.

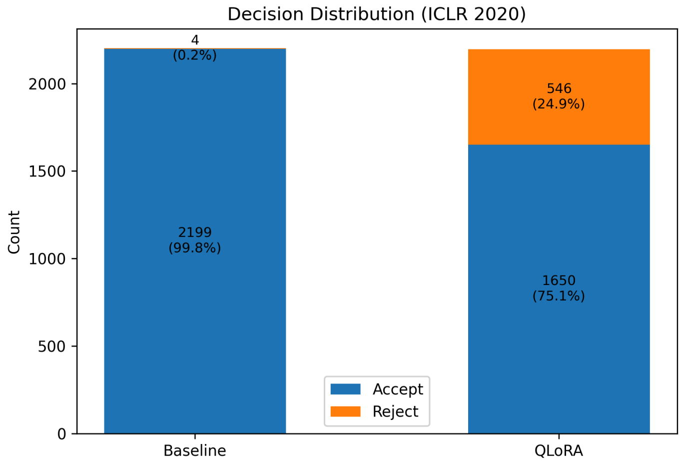


Figure 4. Decision distribution (counts). Note: Baseline N=2203, QLoRA N=2196.

Threats to Validity and Limitations

The underlying data remain noisy and biased: OpenReview texts are heterogeneous, and editorial decisions may reflect venue-specific norms rather than universally shared criteria. The class imbalance in the test set (69% Reject) further complicates interpretation: as noted in Results, both models fall below the accuracy of a naive always-Reject classifier, which sets an important lower bound that automatic metrics alone do not capture. Holding out the 2020 cycle introduces a realistic distribution shift, but it also changes the topical mix of submissions, making it harder to disentangle methodological improvements from topic effects. Different similarity metrics tell different stories: QLoRA improves lexical overlap (Jaccard, ROUGE, chrF) yet the zero-shot backbone still fares better on semantic similarity (SBERT precision/recall/F1), and these automatic metrics are imperfect proxies for human judgment. Moreover, by generating the decision separately from the review, the model never has to justify its verdict within the narrative; linking the verdict to a structured rubric and scoring both the decision and its rationale would provide a stronger test of review quality. External validity is limited: results are reported only on ICLR-style submissions using a compact 3B-parameter backbone; larger models and cross-venue evaluation would be needed to generalise. Finally, the small, skewed dataset and limited GPU budget restrict hyperparameter exploration and may underestimate variance.

Reproducibility Notes

The pipeline is intentionally simple and fully documented. After preprocessing the data — with editorial decisions included only in the completion target, never in the prompt — the data is split into train/validation/test partitions (ICLR 2017–2019 for training and validation, ICLR 2020 for out-of-distribution testing). The zero-shot baseline follows a two-step strategy: one prompt elicits a structured review, a second forces a single-token Accept/Reject verdict. For fine-tuning, QLoRA adapters are trained on the binary labels and the best checkpoint is selected by validation loss. At evaluation time, each test paper is matched strictly by normalised (title, abstract) keys, and the true decision is never exposed in the prompt. The full pipeline — data preprocessing, training, zero-shot baseline and evaluation scripts — is available at the linked repository; all hyperparameters, random seeds and checkpoint-selection criteria are documented in the accompanying notebook.

Conclusion

This study demonstrates that a compact 3B-parameter backbone adapted with QLoRA learns not only the "voice" of a reviewer but, crucially, a more balanced selectivity for Accept vs. Reject decisions. Compared with a fluent yet biased zero-shot baseline, the QLoRA-tuned model improves macro F1 from 0.237 to 0.415 and yields higher lexical-overlap scores, though neither model surpasses a naive always-Reject classifier in overall accuracy — a reminder that per-class metrics are essential on imbalanced data. The baseline retains an edge on semantic-similarity measures such as SBERT, reflecting a trade-off between domain-specific lexical alignment and fluent abstract prose that lightweight adaptation cannot fully resolve. The KL-regularisation experiment clarifies that regularisation is context-dependent: with few, noisy labels and a biased teacher distribution, a KL penalty can over-constrain the very changes that matter, raising validation loss and perplexity without improving selectivity. Overall, the approach remains practical and reproducible, offering immediate utility for early screening and author feedback while leaving final judgement to human reviewers.

References

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Link to the project:

https://drive.google.com/drive/folders/1XQKbqUhGCq4b5yfLhBnxhAM3o07K3kSm?usp=drive_link